v2

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Telematics Voice/Audio Customisation Guide

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2020-06-25 12:22:22 UTC
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Agenda

- TelSDK API: Audio Architecture
- Mapping of Audio Devices from TelSDK to ACDB
- TelSDK API: Voice Call
- TelSDK API: Audio Playback
- TelSDK API: Audio Recording
- TelSDK API: Device Switching
- TelSDK API: DTMF Generation
- TelSDK API: Audio Loopback
- TelSDK API: Documentation and Sample Apps
- TelSDK API: Debugging and Key Log Messages

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TelSDK API

Audio Architecture

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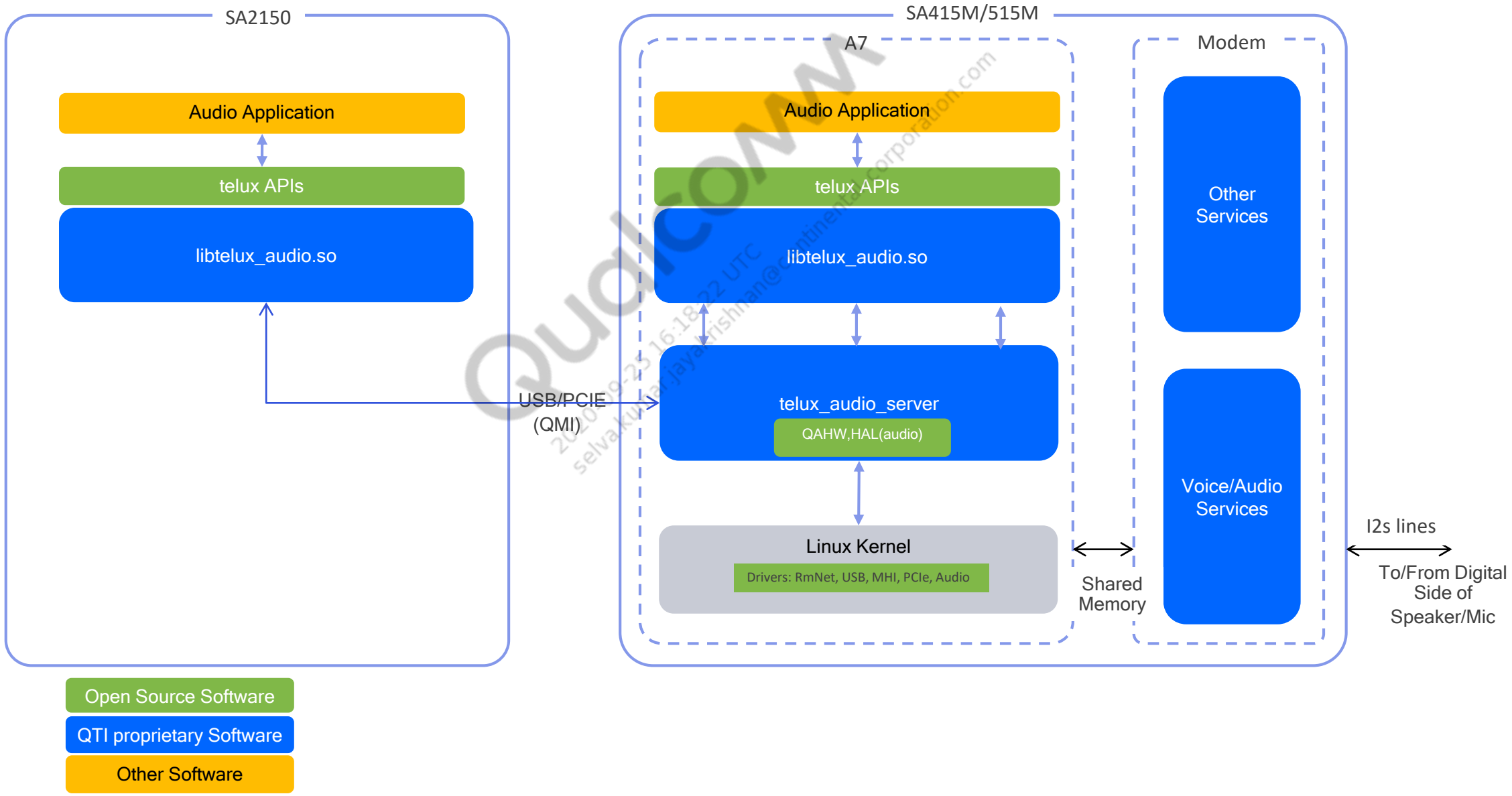
Implementation with External AP Architecture



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Mapping of Audio Devices from TelSDK to ACDB

TelSDK->AudioHAL->ACDB

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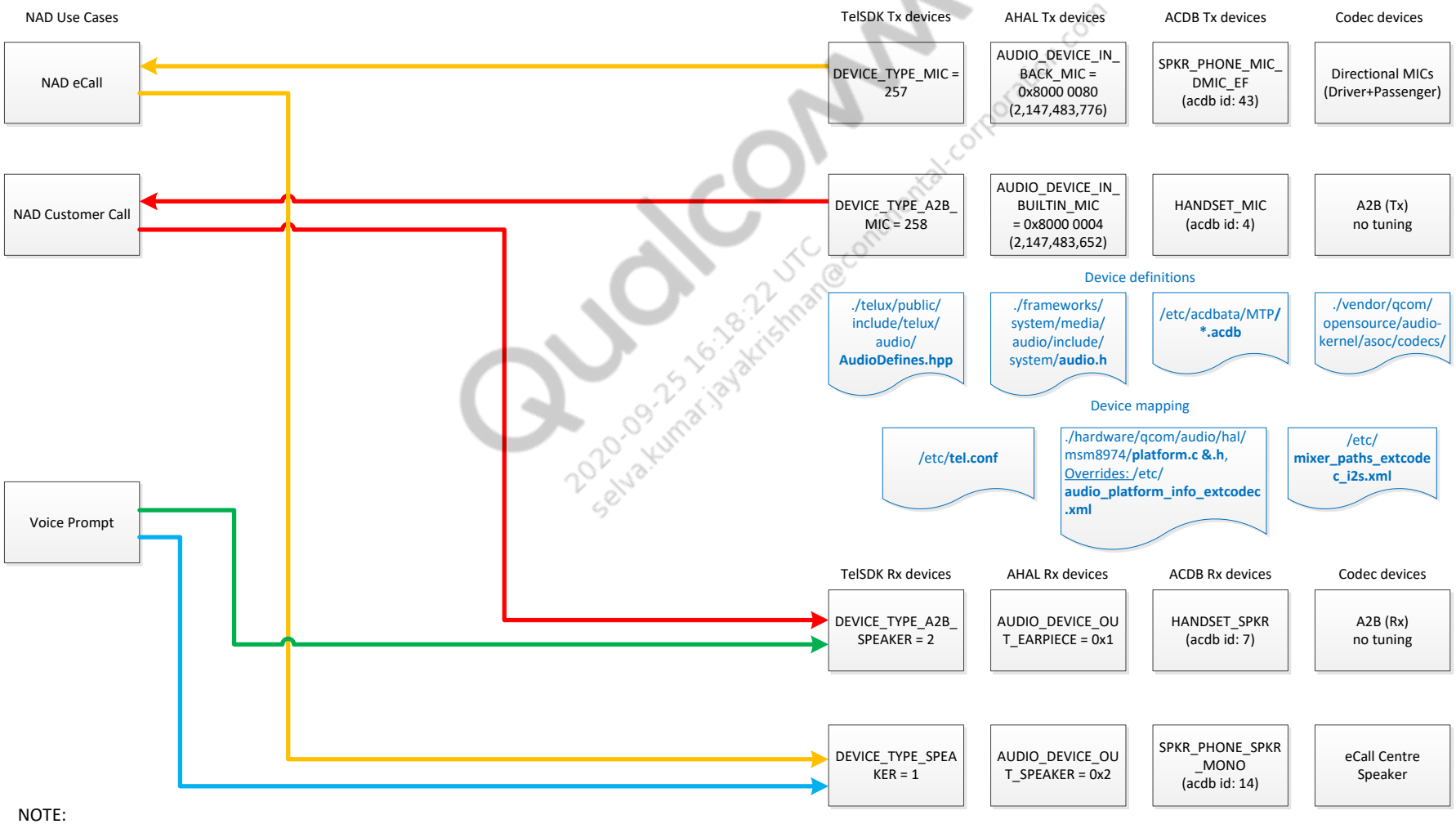
Device Mapping TelSDK-AHAL-ACDB



NOTE:
- For Voice Use Case, only the Rx device Is selected by TelSDK. The matching Tx Device is selected in AHAL.
- For Playback Use Case, the Rx device is selected by TelSDK
- For Record Use Case, the Tx device is selected by TelSDK
- Switch to eCall Centre Speaker AND Directional MICs if A2B speaker fails during NAD Customer Call

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Device mapping TelSDK to Audio HAL

/etc/tel.conf

1. Pull the “tel.conf” configuration file from the target filesystem
adb pull /etc/tel.conf

2. Uncomment and update the four “Device table” lines in the “tel.conf” according to desired device mapping (see diagram on previous slide)

```
# Device table
# NUM_DEVICES specifies the number of device types supported
# DEVICE_TYPE specifies the SDK type of each device (in comma separated values)
# DEVICE_DIR specifies the device direction for each device
#          in order above (in comma separated values)
# AHAL_DEVICE_TYPE specifies the mapped Audio HAL type of each device (in comma separated values)
# Note: The default values provided here are based on QTI's reference design.
NUM_DEVICES=4
DEVICE_TYPE=1,2,257,258
DEVICE_DIR=1,1,2,2
AHAL_DEVICE_TYPE=2,1,2147483776,2147483652
```

3. Push the modified “tel.conf” configuration file back to the target filesystem
adb shell mount -o rw,remount /
adb push tel.conf /etc/.

4. Reboot the target, so that it picks up the new “tel.conf”

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Device mapping Audio HAL to ACDB for dual mic input

/etc/audio_platform_info_extcodec.xml

1. Pull the “audio_platform_info_extcodec.xml” configuration file from the target filesystem
adb pull /etc/audio_platform_info_extcodec.xml

2. Add the following <acdb_ids> lines at the bottom of the “audio_platform_info_extcodec.xml”.
This is to override the 'acdb_device_table()' audio hal to acdb device mappings in "hardware/qcom/audio/hal/msm8974/platform.c“
Basically, this will replace the mapping to:
single mic acdb device SPKR_PHONE_MIC (11)
with mapping to:
dual mic acdb device SPKR_PHONE_MIC_DMIC_EF (43)

```
  <acdb_ids>
    <device name="SND_DEVICE_OUT_VOICE_SPEAKER" acdb_id="14" />
    <device name="SND_DEVICE_IN_VOICE_SPEAKER_MIC" acdb_id="43" />
    <device name="SND_DEVICE_OUT_VOICE_HANDSET" acdb_id="7" />
    <device name="SND_DEVICE_IN_HANDSET_MIC" acdb_id="4" />      ← OPTIONAL: acdb_id =“41” could be used instead of “4” if A2B Tx is also a dual mic input
  </acdb_ids>

</audio_platform_info_i2s>
```

3. Push the modified “audio_platform_info_extcodec.xml” configuration file back to the target filesystem
adb shell mount -o rw,remount /
adb push audio_platform_info_extcodec.xml /etc/.

4. Reboot the target, so that it picks up the new “audio_platform_info_extcodec.xml”

NOTE:
Since we are reusing existing devices in ACDB, no changes are required in those parts for this device mapping to function.
However, the size of the ACDB files can be reduced by removing all of the device pairs and devices which are not used.

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Codec-specific customisation

/etc/mixer_paths_extcodec_i2s.xml

1. Pull the “mixer_paths_extcodec_i2s.xml” configuration file from the target filesystem
adb pull /etc/ mixer_paths_extcodec_i2s.xml

2. Customise codec-specific mixer settings in “mixer_paths_extcodec_i2s.xml”.
a) To ensure that the path through the 3rd party external codec to set correctly (relevant mixer controls to be filled in for each sound device according to 3rd party codec driver)
b) To define the required number of channels going to and from the I2S (see highlighted lines below according to your requirements)

```
<!-- These are actual sound device specific mixer settings -->
  <path name="speaker">
    <ctl name="MI2S_RX Channels" value="One" />
  </path>

  <path name="speaker-mic">
    <ctl name="MI2S_TX Channels" value="Two" />
  </path>

  <path name="handset">
    <ctl name="MI2S_RX Channels" value="One" />
  </path>

  <path name="handset-mic">
    <ctl name="MI2S_TX Channels" value="One" />
  </path>

  <path name="voice-speaker">
    <path name="speaker" />
  </path>

  <path name="voice-speaker-mic">
    <path name="speaker-mic" />
  </path>

  <path name="voice-handset">
    <path name="handset" />
  </path>

  <path name="voice-handset-mic">
    <path name="handset-mic" />
  </path>
```

3. Push the modified “mixer_paths_extcodec_i2s.xml” configuration file back to the target filesystem
adb shell mount -o rw,remount /
adb push mixer_paths_extcodec_i2s.xml /etc/.

4. Reboot the target, so that it picks up the new “mixer_paths_extcodec_i2s.xml”

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Testing the new device mapping on the target (1)

```
adb shell
audio_console_app

Elapsed Time for Audio Subsystems to ready : 0.0055951s
-----------------------------------------------
            Audio console app - SDK v1.33.1
-----------------------------------------------
  1 - Voice Call
  2 - Playback
  3 - Capture
  4 - Loopback
  5 - Tone
  6 - TransCode

  ? / h - help
  q / 0 - exit
-----------------------------------------------
audio> 1
```

```
-----------------------------------------------
            Voice Menu
-----------------------------------------------
  1 - Create Stream
  2 - Delete Stream
  3 - Get Device
  4 - Set Device
  5 - Get Volume
  6 - Set Volume
  7 - Get Mute Status
  8 - Set Mute
  9 - Start Audio
  10 - Stop Audio
  11 - Start Dtmf Tone
  12 - Stop Dtmf Tone
  13 - Register Listener
  14 - Deregister Listener

  ? / h - help
  q / 0 - exit
-----------------------------------------------
voice> 1
Enter Modem Id: 1
Enter Sample Rate (16000 32000 48000) :16000
Enter channel mask (1 for left, 2 for right, 3 for both): 1 (single mic) OR 3 (dual mic)
Enter no. of devices : 1
Enter device type (1 for speaker, 257 for microphone): 1
Request to create stream sent
Audio Stream is Created
Audio Voice Stream is Created
voice> 9
Request to start Audio sent
Audio Stream is Started
```

Repeat this test for the two device pairs (or select 4-Set Device option to dynamically change device during an ongoing use case):

1) Speaker 1, Mic 257 for “SPKR_PHONE_MIC_DMIC_EF&SPKR_PHONE_SPKR_MONO”
2) Speaker 2, Mic 258 for “HANDSET_MIC&HANDSET_SPKR”

Only the Rx (speaker) device needs to be entered.
The Tx (microphone) device will be automatically selected based on the Rx device by Audio HAL
Note: Sample Rate and Channel are entered but not used in the console app, since these are decided by the cellular network

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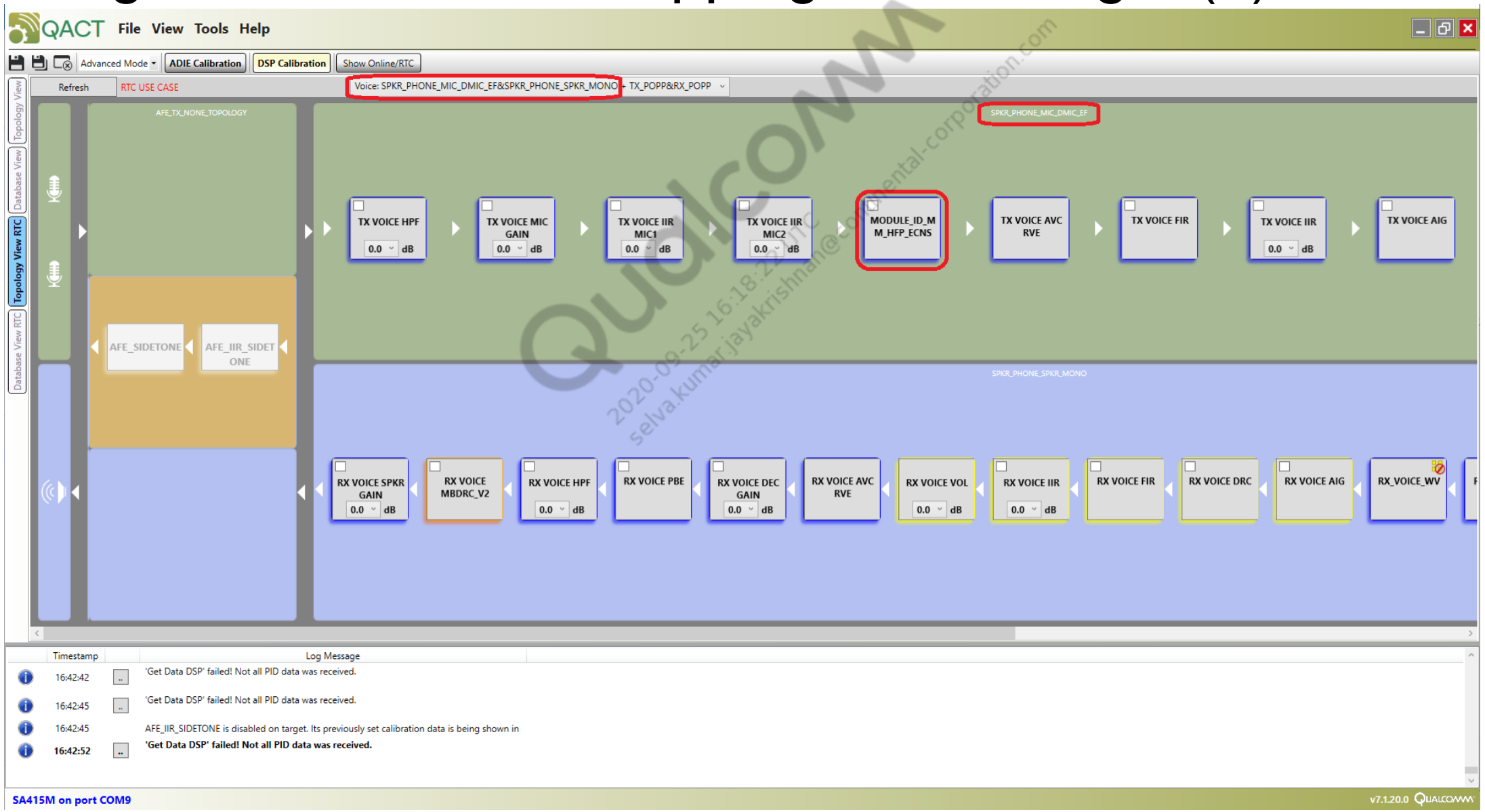
Testing the new device mapping on the target (2)



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Dynamic switching of device during ongoing use case

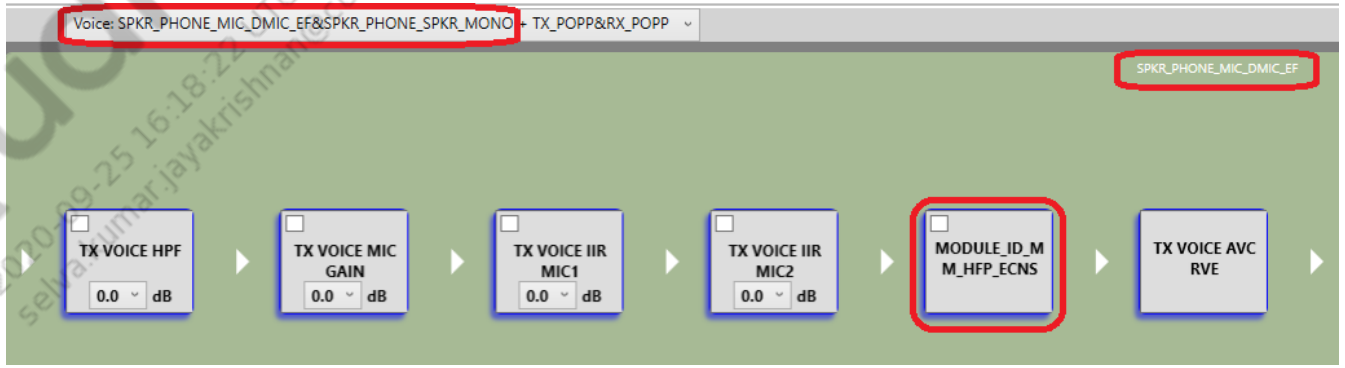
Prerequisites:
1. Enter the adb shell and execute the audio_console_app
2. Start a voice call by creating stream, and starting audio
3. Set device by entering “4” from the Voice Menu….

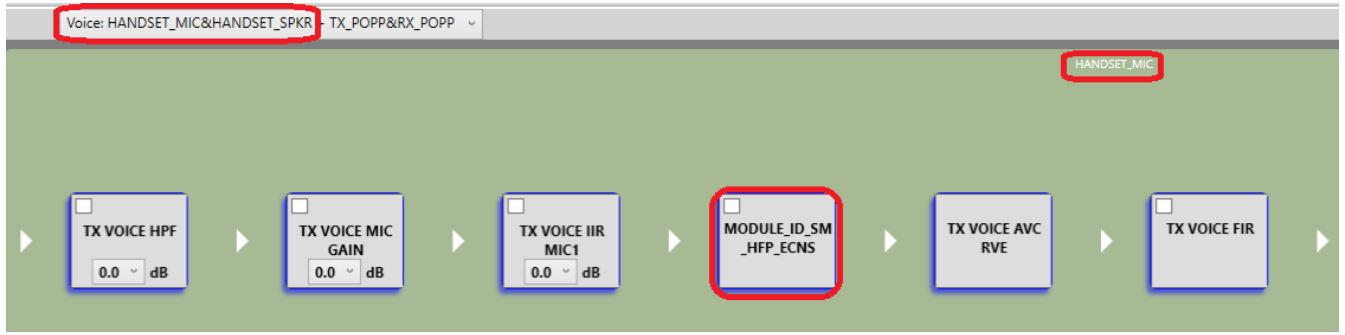
```
-----------------------------------------------
                Voice Menu
-----------------------------------------------
  1 - Create Stream
  2 - Delete Stream
  3 - Get Device
  4 - Set Device
  5 - Get Volume
  6 - Set Volume
  7 - Get Mute Status
  8 - Set Mute
  9 - Start Audio
  10 - Stop Audio
  11 - Start Dtmf Tone
  12 - Stop Dtmf Tone
  13 - Register Listener
  14 - Deregister Listener

  ? / h - help
  q / 0 - exit
 -----------------------------------------------
 voice> 4
Enter no. of devices : 1
Enter device type (1 for speaker, 257 for microphone): 2
Request to set device sent
set stream device succeeded.
```

4. Test for speaker=1 and speaker=2
5. Check that correct speaker and mic have been selected in QACT RTC mode

NOTE: The microphone will be automatically selected by Audio HAL based on the speaker.





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Device definitions for customer app

./telux/public/include/telux/audio/AudioDefines.hpp

Add one new mic and one new speaker to “DeviceType” enumeration structure according to desired device mapping (see diagram on previous slide)

```
enum DeviceType {
DEVICE_TYPE_NONE = -1,
/* output devices */
DEVICE_TYPE_SPEAKER = 1,
DEVICE_TYPE_A2B_SPEAKER = 2,
 /* input devices */
DEVICE_TYPE_MIC = 257,
DEVICE_TYPE_A2B_MIC = 258,
};
```

These enumerated values can then be used in the customer application to select the correct speaker and microphone for a given use case.

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Custom ACDB files location

Changing the acdbdata path from /etc to a different directory

Steps are as follows (example given is a new location of “/data/acdbdata/MTP”):

1) Set ACDB_BIN_PATH variable in Makefile.am to point to custom acdb file path:
./audio/mm-audio-cal/audio-acdb-util/acdb-loader/src/Makefile.am
-libacdbloader_la_CFLAGS = $(AM_CFLAGS) -DUSE_GLIB @GLIB_CFLAGS@ -D WCD9320_ANC_BIN_PATH=\"/data/misc/audio/wcd9320_anc.bin\" -D MBHC_BIN_PATH=\"/data/misc/audio/mbhc.bin\" -D MAD_BIN_PATH=\"/data/misc/audio/wcd9320_mad_audio.bin\" -D ETC_ROOT_PATH=\"/etc/\" -D ACDB_BIN_PATH=\"/etc/acdbdata/\" -D DEFAULT_BOARD=\"MTP\"
+libacdbloader_la_CFLAGS = $(AM_CFLAGS) -DUSE_GLIB @GLIB_CFLAGS@ -D WCD9320_ANC_BIN_PATH=\"/data/misc/audio/wcd9320_anc.bin\" -D MBHC_BIN_PATH=\"/data/misc/audio/mbhc.bin\" -D MAD_BIN_PATH=\"/data/misc/audio/wcd9320_mad_audio.bin\" -D ETC_ROOT_PATH=\"/etc/\" -D ACDB_BIN_PATH=\"/data/acdbdata/\" -D DEFAULT_BOARD=\"MTP\"

2) SELinux permission should be added in audio_hal.fc for that path.
./poky/meta-qti-sepolicy/recipes-security/sepolicy/common/system/audio_hal.fc
+/data/acdbdata/MTP(/.*)? gen_context(system_u:object_r:audio_hal_etc_acdb_t,s0)

3) ACDB files, workspace file, and ACDB delta files should be manually copied/pushed into custom ACDB path in the target file system for this to work

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TelSDK API

Voice Call

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TelSDK API

To Enable Voice Call Paths and Streams (Uplink and Downlink Directions)

Action	Legacy API	TelSDK API
Enable Voice Call Path	amix 'VoiceMMode1_Tx Mixer PRI_TX_MMode1' 1 amix 'PRI_MI2S_RX_Voice Mixer VoiceMMode1' 1 arec -Dhw:0,2 -P | aplay -Dhw:0,2 -P	IAudioManager::createStream • type(VOICE_CALL) • slotId(DEFAULT_SLOT_ID) • format(PCM_16BIT_SIGNED) • sampleRate(16000) • channelTypeMask(LEFT) • deviceTypes(DEVICE_TYPE_SPEAKER) IAudioVoiceStream::startAudio

Notes:
1 - Create Stream in “audio_console_app” Voice menu, followed by 9 - Start Audio

For Single SIM, voice is enabled based on DEFAULT_SLOT_ID which currently maps to SLOT_ID_1.
For Dual SIM, voice would be enabled based on selected SLOT_ID_1/SLOT_ID_2.

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TelSDK API

To Disable Voice Call Paths and Streams (Uplink and Downlink Directions)

Action	Legacy API	TelSDK API
Disable Voice Call Path	(kill associated arec and aplay processes if running in the background) amix 'VoiceMMode1_Tx Mixer PRI_TX_MMode1' 0 amix 'PRI_MI2S_RX_Voice Mixer VoiceMMode1' 0	IAudioVoiceStream::stopAudio IAudioManager::deleteStream
Note: 10 - Stop Audio in “audio_console_app” Voice menu, followed by 2 - Delete Stream		

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TelSDK API

To Mute Uplink and Downlink Voice Streams in DSP Audio Subsystem

Action	Legacy API	TelSDK API
Mute Voice Call Path	amix 'Voice Tx Device Mute' 1 297816064 0 amix 'Voice Rx Device Mute' 1 297816064 0	IAudioVoiceStream::setMute • enable(TRUE) • dir(TX) IAudioVoiceStream::setMute • enable(TRUE) • dir(RX)
Note: 8 - Set Mute in “audio_console_app” Voice menu		

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TelSDK API

To Unmute Uplink and Downlink Voice Streams in DSP Audio Subsystem

Action	Legacy API	TelSDK API
Unmute Voice Call Path	amix 'Voice Tx Device Mute' 0 297816064 0 amix 'Voice Rx Device Mute' 0 297816064 0	IAudioVoiceStream::setMute • enable(FALSE) • dir(TX) IAudioVoiceStream::setMute • enable(FALSE) • dir(RX)
Note; 8 - Set Mute in “audio_console_app” Voice menu		

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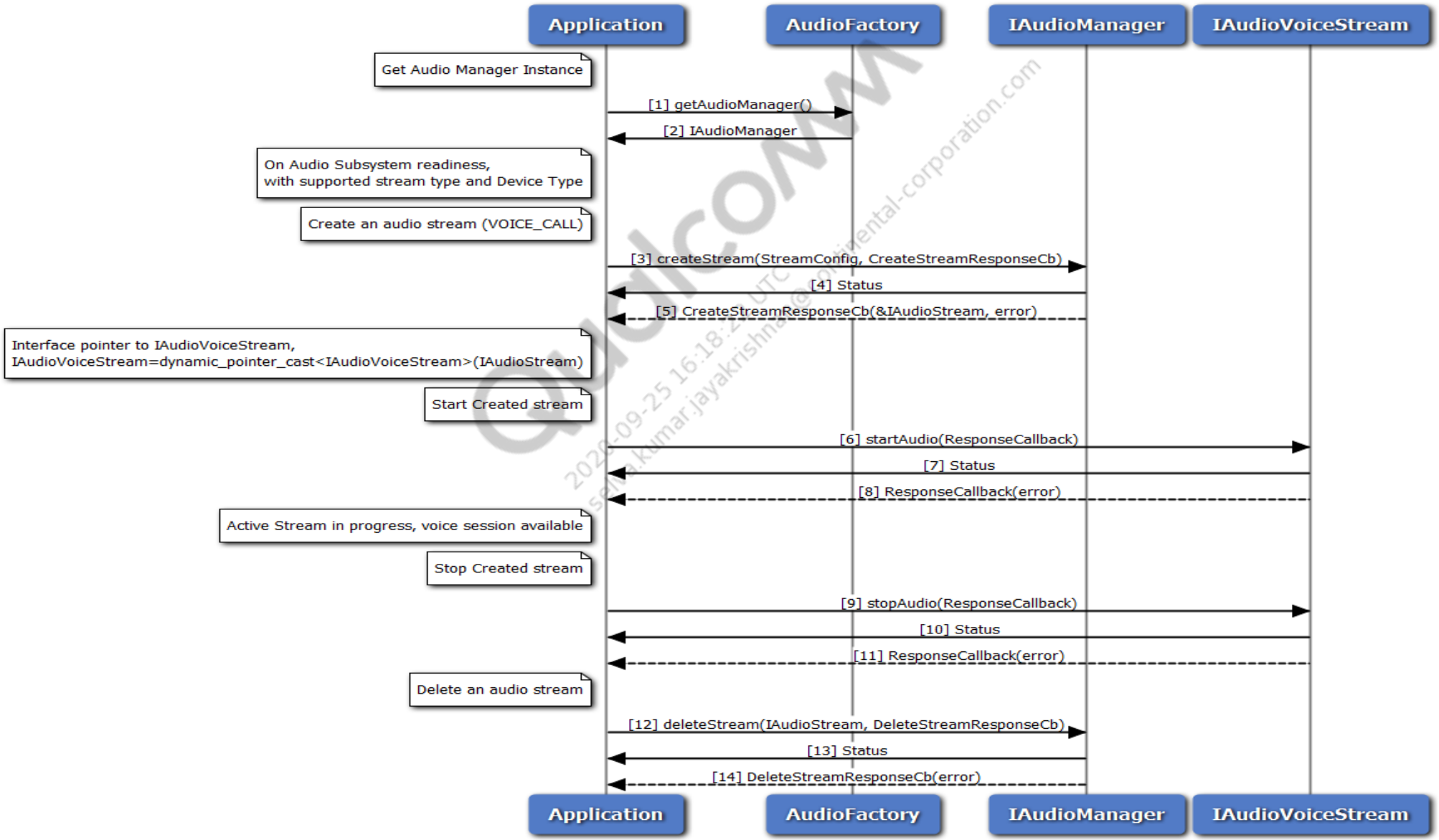
TelSDK Voice Session Call Flow (Speaker/Mic)



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TelSDK API

Audio Playback

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TelSDK API

To Enable Prompt path (PCM Playback):

Action	Legacy API	TelSDK API
Enable Prompt path (Playback)	amix 'PRI_MI2S_RX Audio Mixer MultiMedia1' 1 aplay -Dhw:0,0 <path-to-prompt-file.wav>	IAudioManager::createStream • type(PLAY) • slotId(DEFAULT_SLOT_ID) • format(PCM_16BIT_SIGNED) • sampleRate(X) • channelTypeMask(Y) • deviceTypes(DEVICE_TYPE_SPEAKER) Allocate Stream Buffers, which would provide PCM samples from Prompt file(*.wav) to Playback Stream IAudioPlayStream::getStreamBuffer Using Allocated Stream Buffers, read PCM samples from Prompt file(*.wav) and Write to Playback Stream till End Of File reached. IAudioPlayStream::write

Notes:
1 - Create Stream in “audio_console_app” Playback menu, followed by 9 - Start Play
Typical Create Stream options:
Enter Modem Id: 1
Enter Sample Rate (16000 32000 48000) : 48000
Enter channel mask (1 for left, 2 for right, 3 for both): 1 (if playing mono file) OR 3 (if playing stereo file)
Please Select Audio Format : 1->PCM, 2->AMRWB+, 3->AMRNB, 4->AMRWB : 1
Enter File name with path : /data/test.wav
Enter no. of devices : 1
Enter device type (1 for speaker, 257 for microphone): 1
Enter voice path type (0 for None, 1 for RX, 2 for TX, 3 for BOTH): 0

Application has responsibility of Parsing Wav file header to get Stream Property (X, Y) and reading PCM samples from Prompt file(*.wav), providing read PCM samples to Playback Stream.

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TelSDK API

To Disable Prompt path (PCM Playback):

Action	Legacy API	TelSDK API
Disable Prompt path (Playback)	amix 'PRI_MI2S_RX Audio Mixer MultiMedia1' 0	When Application Decide to Stop Playback Stream, need to invoke IAudioManager::deleteStream

Note:
10 - Stop Play in “audio_console_app” Playback menu (if not finishing playing back), followed by 2 - Delete Stream

Application has responsibility of releasing file resource specific to Prompt file(*.wav).

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TelSDK API

To Enable Prompt path (AMR Playback):

Action	Legacy API	TelSDK API
Enable Prompt path (AMR Playback)	Not applicable	IAudioManager::createStream • type(PLAY) • slotId(DEFAULT_SLOT_ID) • format(AMRWB_PLUS | AMRNB | AMRWB) • sampleRate(X) • channelTypeMask(Y) • deviceTypes(DEVICE_TYPE_SPEAKER) Allocate Stream Buffers, which would provide PCM samples from Prompt file(*.raw) to Playback Stream IAudioPlayStream::getStreamBuffer Using Allocated Stream Buffers, read PCM samples from Prompt file(*.raw) and Write to Playback Stream till End Of File reached. IAudioPlayStream::write

Notes:
1 - Create Stream in “audio_console_app” Playback menu, followed by 9 - Start Play
Typical Create Stream options:
Enter Modem Id: 1
Enter Sample Rate (16000 32000 48000) : 48000
Enter channel mask (1 for left, 2 for right, 3 for both): 1 (if playing mono file) OR 3 (if playing stereo file)
Please Select Audio Format : 1->PCM, 2->AMRWB+, 3->AMRNB, 4->AMRWB : 2
Enter File name with path : /data/amr_wb_plus.raw (file should contain raw data, with no header)
Enter no. of devices : 1
Enter device type (1 for speaker, 257 for microphone): 1
Enter voice path type (0 for None, 1 for RX, 2 for TX, 3 for BOTH): 0

Application has responsibility of Parsing file header to get Stream Property (X, Y) and reading raw samples from Prompt file(*.raw), providing read raw samples to Playback Stream.

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TelSDK API

To Disable Prompt path (AMR Playback):

Action	Legacy API	TelSDK API
Disable Prompt path (AMR Playback)	Not applicable	When Application Decide to Stop Playback Stream, need to invoke IAudioManager::deleteStream

Note:
10 - Stop Play in “audio_console_app” Playback menu (only if not finishing playing back), followed by 2 - Delete Stream

Application has responsibility of releasing file resource specific to Prompt file(*.raw).

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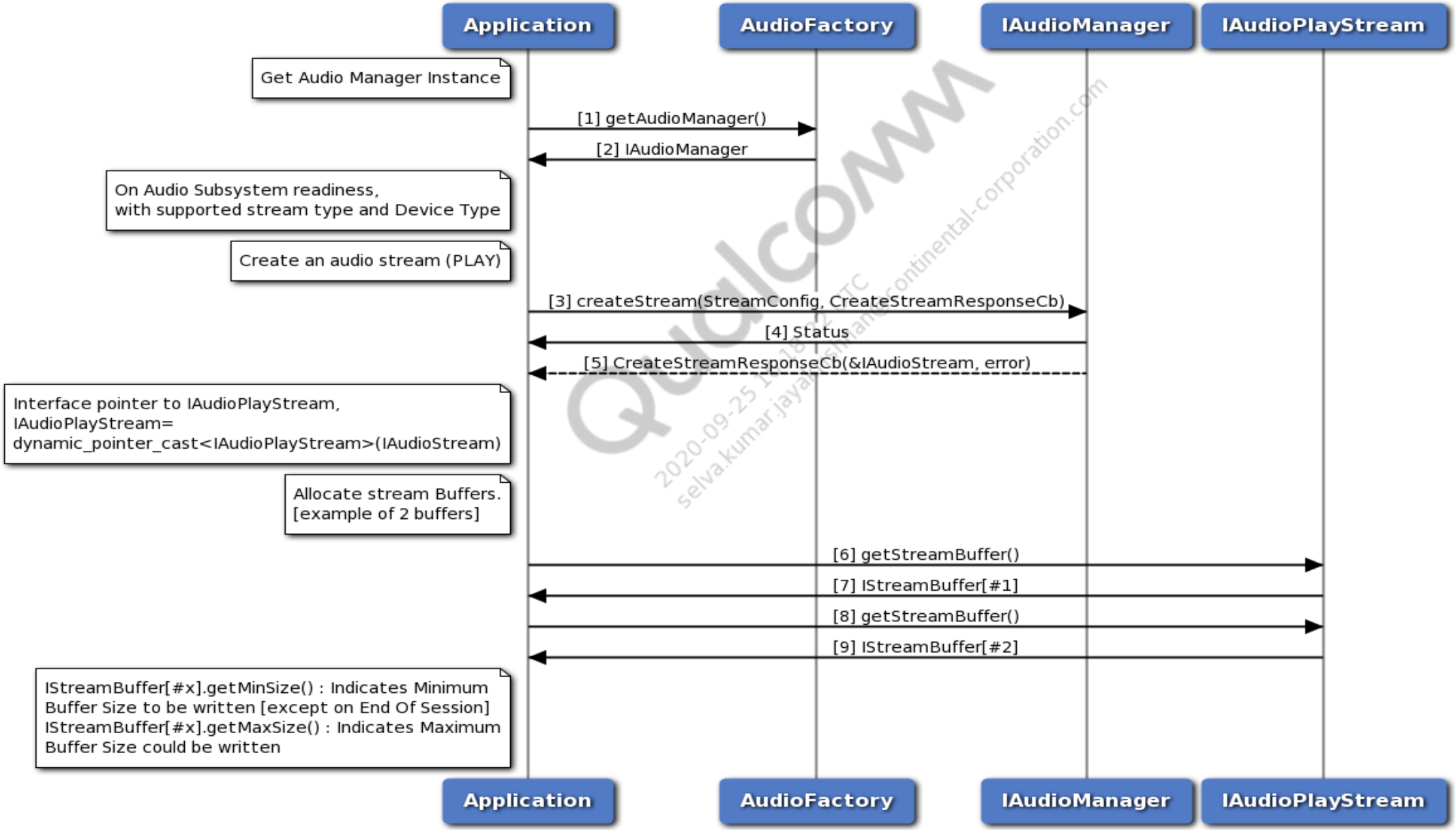
TelSDK Audio Playback Call Flow (Speaker)



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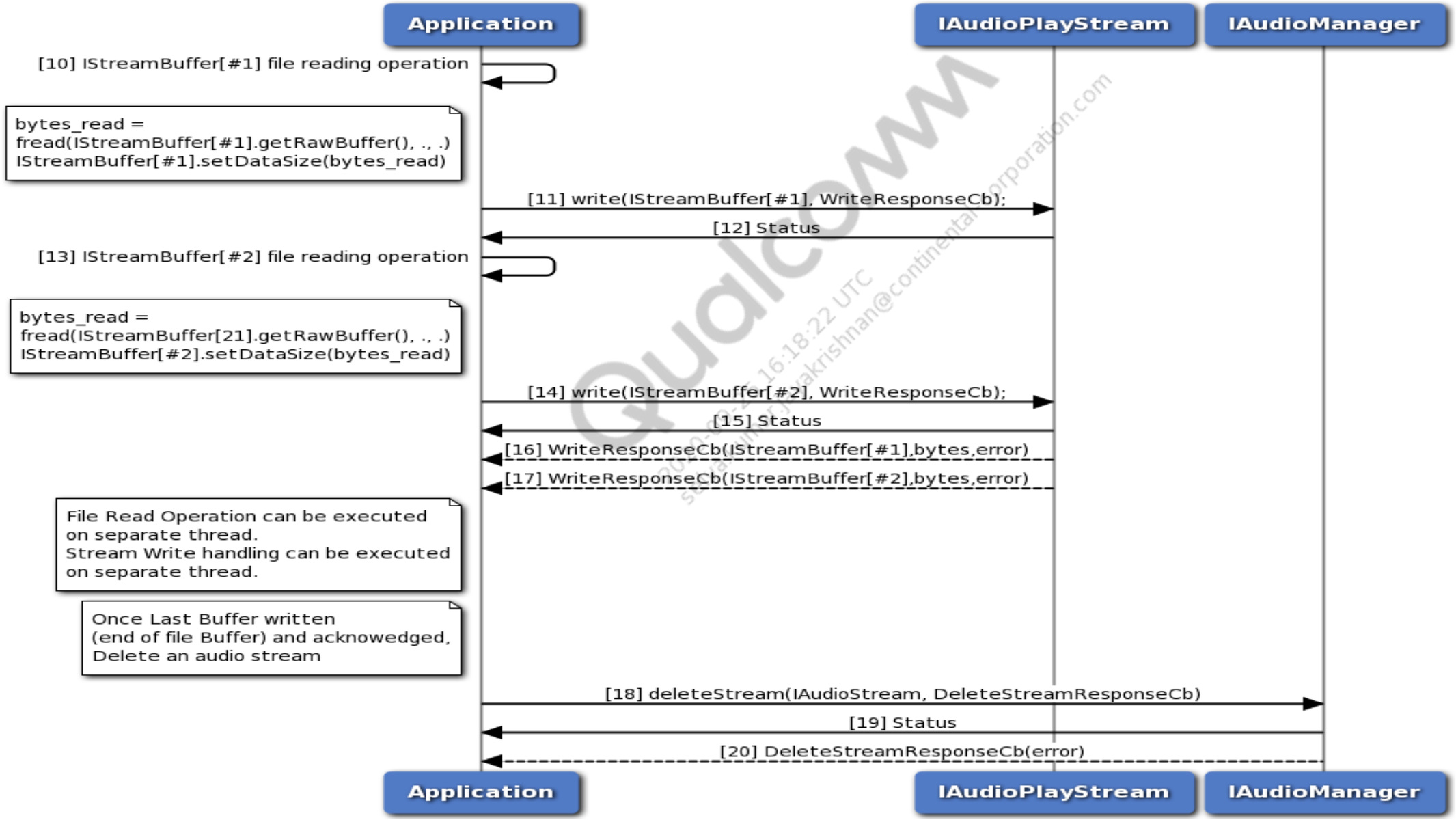
TelSDK Audio Playback Call Flow (Speaker)…



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TelSDK API

Audio Recording

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TelSDK API

To Enable Prompt path (PCM Capture):

Action	Legacy API	TelSDK API
Enable Prompt path (Capture)	amix 'MultiMedia1 Mixer PRI_MI2S_TX' 1 arec -Dhw:0,0 <path-to-capture-file.wav>	IAudioManager::createStream • type(CAPTURE) • slotId(DEFAULT_SLOT_ID) • format(PCM_16BIT_SIGNED) • sampleRate(X') • channelTypeMask(Y') • deviceTypes(DEVICE_TYPE_MIC) Allocate Stream Buffers, which would provide PCM samples from Capture Stream to Prompt file(*.wav) IAudioCaptureStream::getStreamBuffer Using Allocated Stream Buffers, read PCM samples from Capture Stream and Write to Prompt file(*.wav) till user decision to stop. IAudioCaptureStream::read

Note:
1 - Create Stream in "audio_console_app" Capture menu, followed by 9 - Start Capture

Application has responsibility of Creating Wav file header with Stream Property (X', Y') and reading PCM samples from Capture Stream, providing read PCM samples to Prompt file(*.wav).

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TelSDK API

To Disable Prompt path (PCM Capture):

Action	Legacy API	TelSDK API
Disable Prompt path (Capture)	amix 'MultiMedia1 Mixer PRI_MI2S_TX' 0	When Application Decide to Stop Capture Stream, need to invoke IAudioManager::deleteStream

Note:
10 - Stop Capture in “audio_console_app” Capture menu (if not finishing capture), followed by 2 - Delete Stream

Application has responsibility of releasing file resource specific to Prompt file(*.wav).

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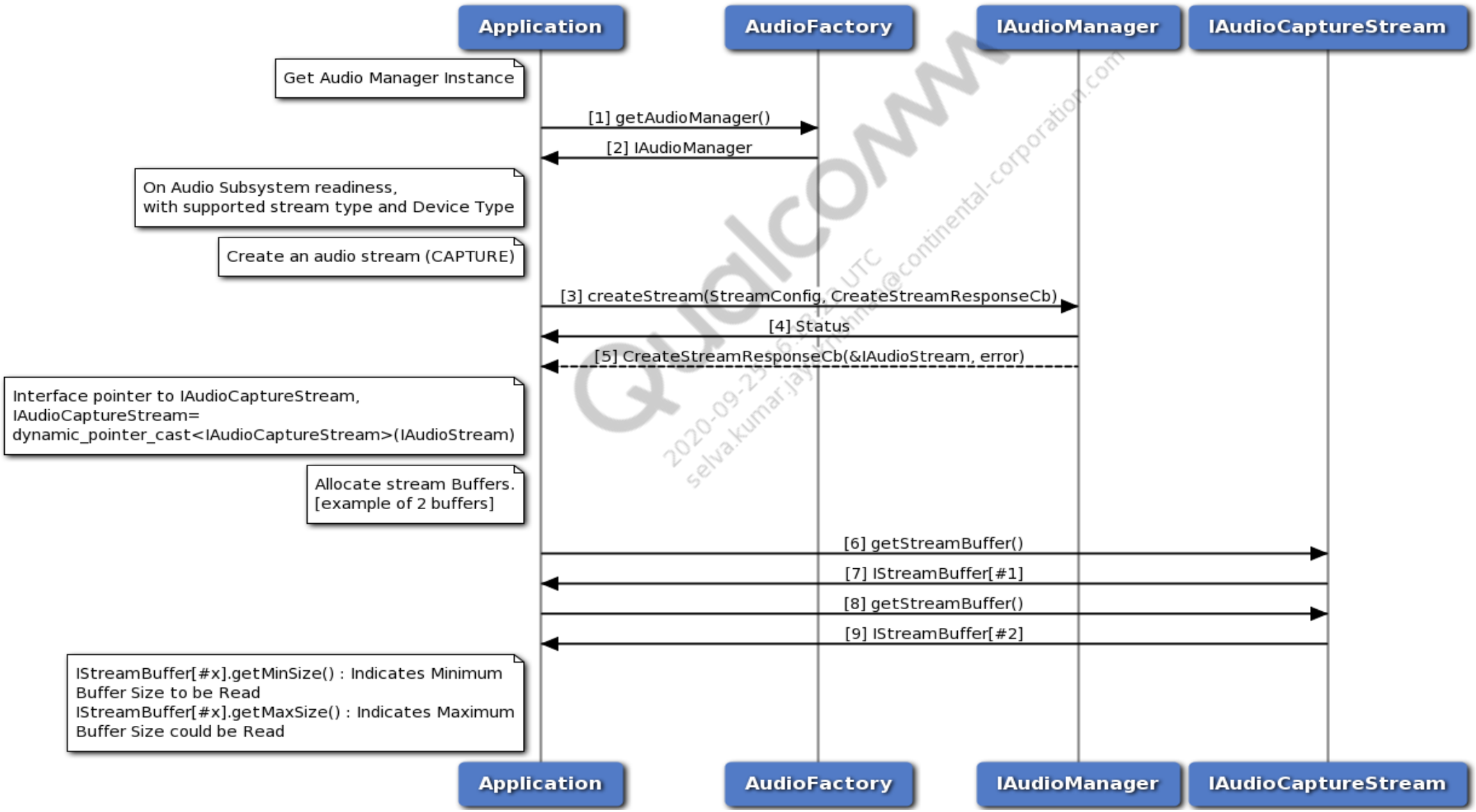
TelSDK Audio Capture Call Flow (Mic)



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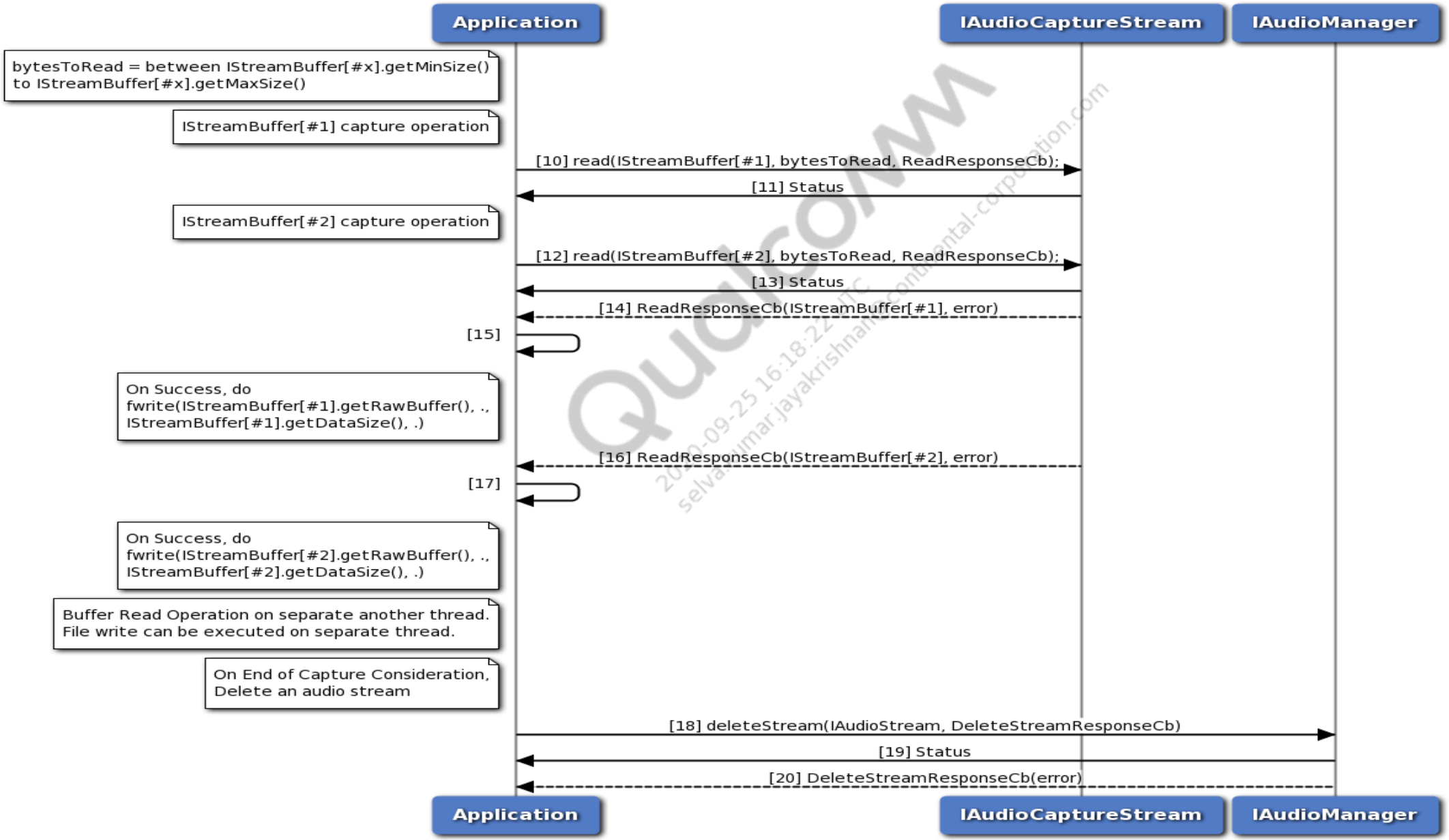
TelSDK Audio Capture Call Flow (Mic)…



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TelSDK API

Device switching

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Device Switching:

Action	Legacy API	TelSDK API
Switch Device	Previously done using ALSA UCM application	IAudioStream::setDevice • devices - Devices list • callback - callback to get the response of setDevice

Note:
4 - Set Device in “audio_console_app” Voice menu, during an ongoing voice call

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TelSDK API

DTMF Generation

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TelSDK API

DTMF Generation (voice downlink)

Action	Legacy API	TelSDK API
Play DTMF to local speaker (during voice call)		IAudioVoiceStream::playDtmfTone • dtmfTone - DTMF tone properties • duration - Duration (in milliseconds) for which the tone needs to be played. The constant infiniteDtmfDuration(=0xFFFF) represents infinite duration • gain - DTMF tone gain (in Q13 format) • callback - callback to get the response of playDtmfTone

Note:
11 - Start Dtmf Tone in “audio_console_app” Voice menu, during an ongoing voice call

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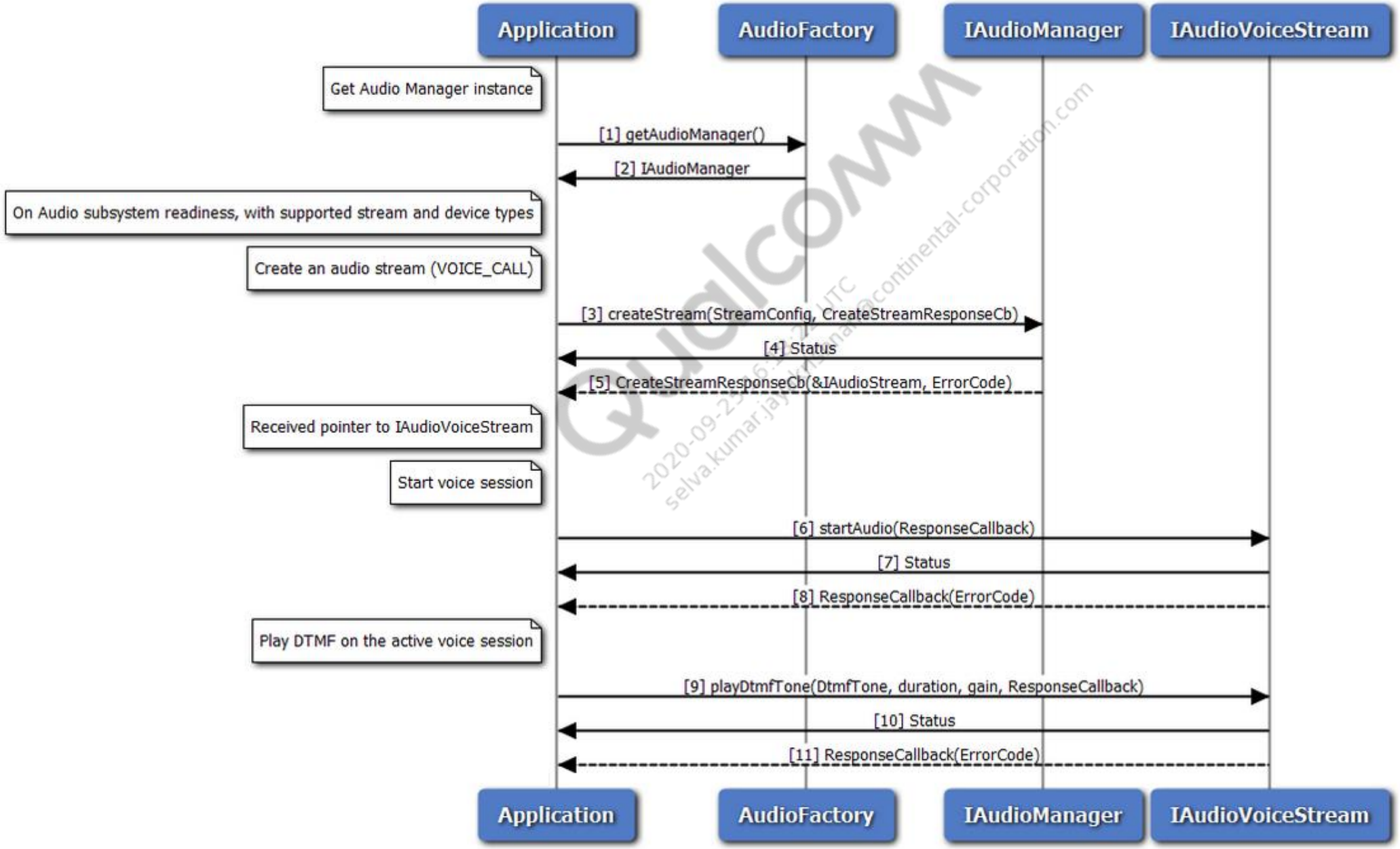
TelSDK Play DTMF Tone Call Flow



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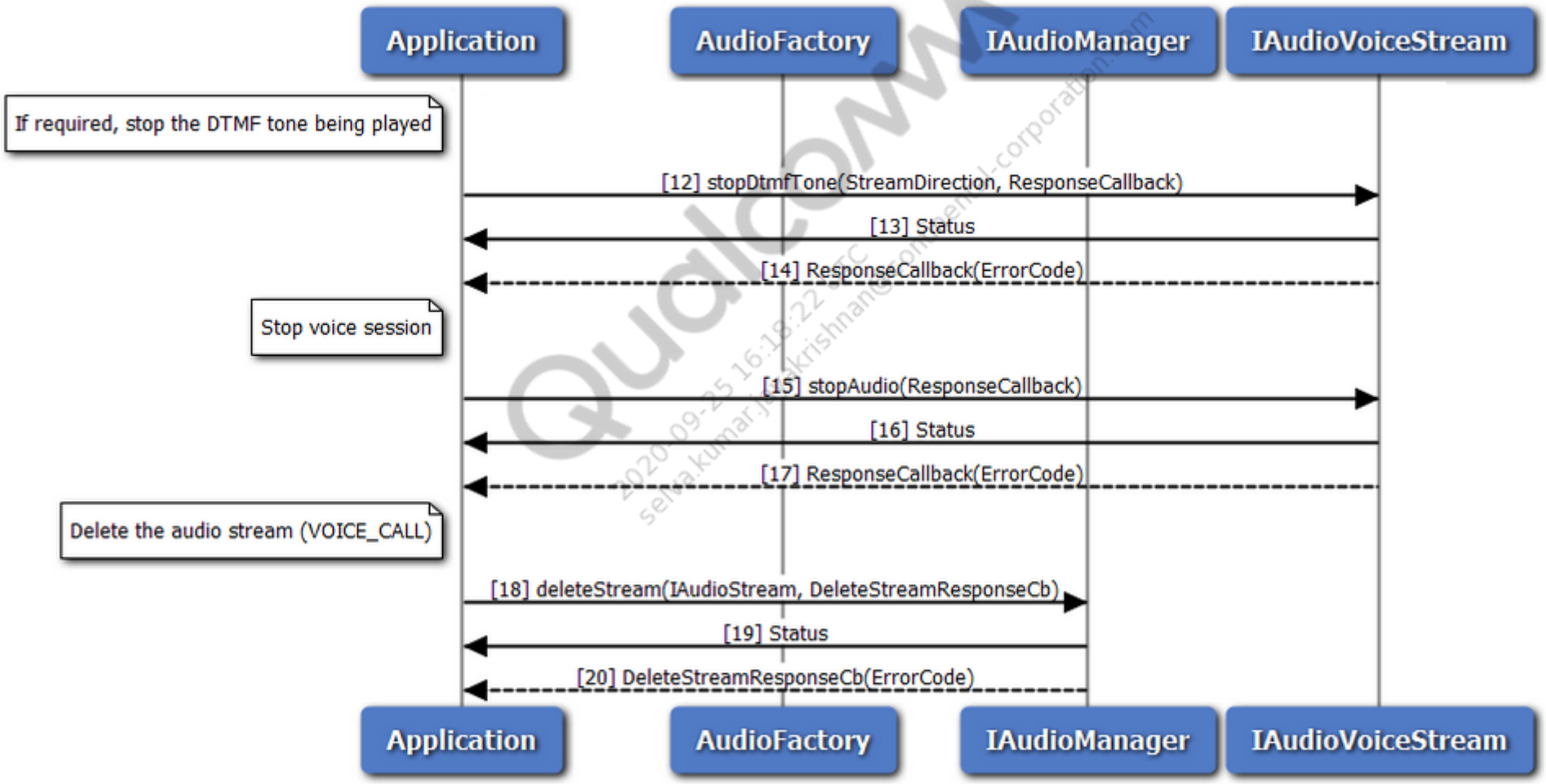
TelSDK Play DTMF Tone Call Flow…



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TelSDK API

DTMF Generation (voice uplink)

Action	Legacy API	TelSDK API
Play DTMF on uplink (during voice call)	QMI_VOICE_START_CONT_DTMF QMI_VOICE_STOP_CONT_DTMF	ICall::playDtmfTone • tone - a single character with one of 12 values: 0-9, *, #. • callback - Optional callback pointer to get the result of playDtmfTones function Play a DTMF tone and stop it. The interval for which the tone is played is dependent on the system implementation. If continuous DTMF tone is playing, it will be stopped.

Note:
12 - Play_Dtmf_Tone in “telsdk_console_app” Dialer menu, during an ongoing voice call

'ICall::playDtmfTone()' allows you to play a DTMF tone for a specific amount of time, rather than continuous, so it calls QMI_VOICE_START_CONT_DTMF, then QMI_VOICE_STOP_CONT_DTMF after a specified time.

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TelSDK API

Audio Loopback

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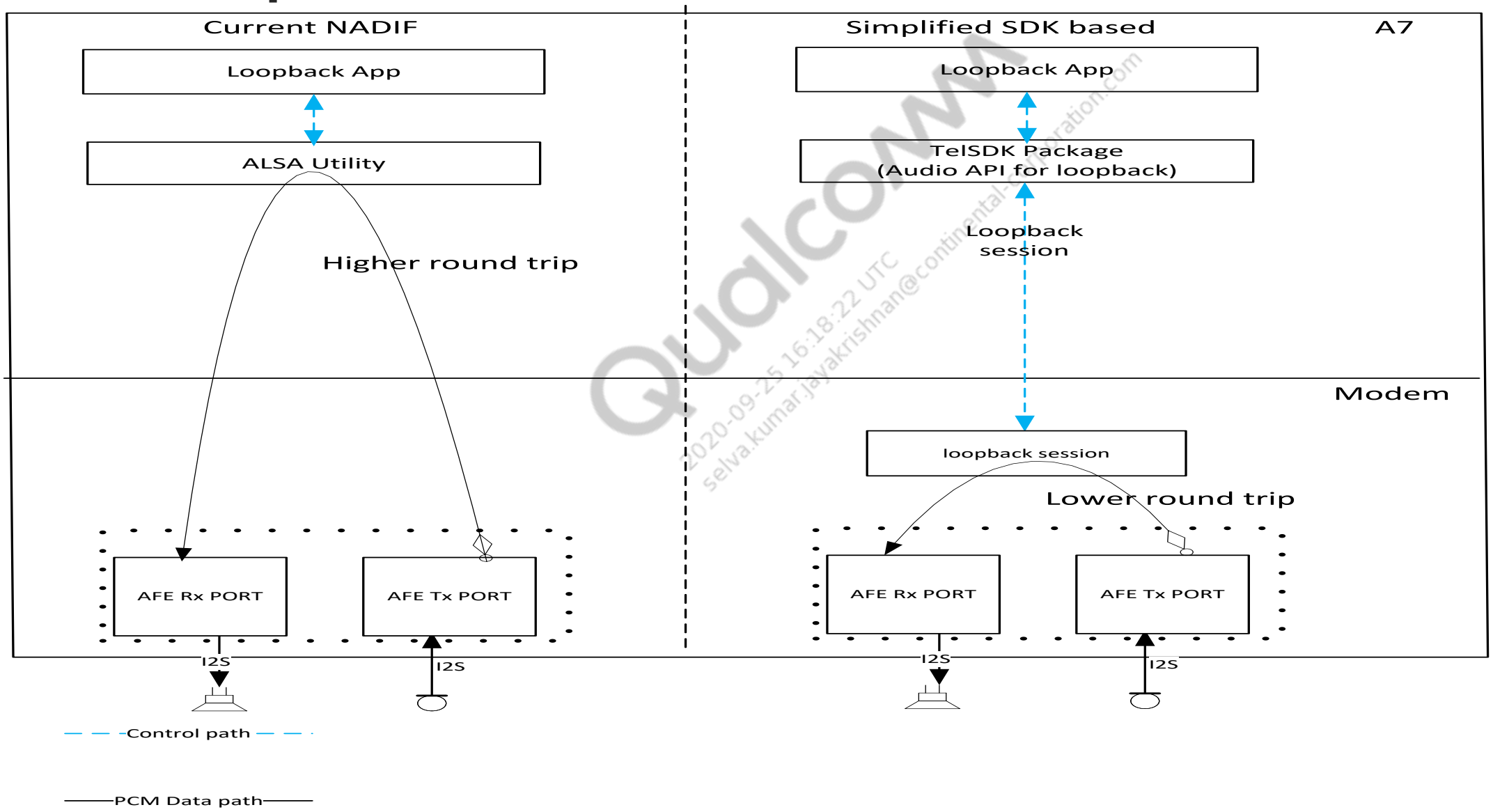
Audio Loopback – ALSA vs AFE



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TelSDK API

To Enable Loopback:

Action	Legacy API	TelSDK API
Enable Loopback	arec -Dhw:0,0 | aplay -Dhw:0,0	IAudioManager::createStream • type(LOOPBACK) • slotId(DEFAULT_SLOT_ID) • format(PCM_16BIT_SIGNED) • sampleRate(X) • channelTypeMask(Y) • deviceTypes(DEVICE_TYPE_SPEAKER, DEVICE_TYPE_MIC) IAudioLoopbackStream::startLoopback

Note:
1 - Create Stream in “audio_console_app” Loopback menu, followed by 9 - Start Loopback

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TelSDK API

To Disable Loopback:

Action	Legacy API	TelSDK API
Disable Loopback	(Kill arec and aplay processes)	IAudioLoopbackStream::stopLoopback IAudioManager::deleteStream

Note:
10 - Stop Loopback in “audio_console_app” Loopback menu, followed by 2 - Delete Stream

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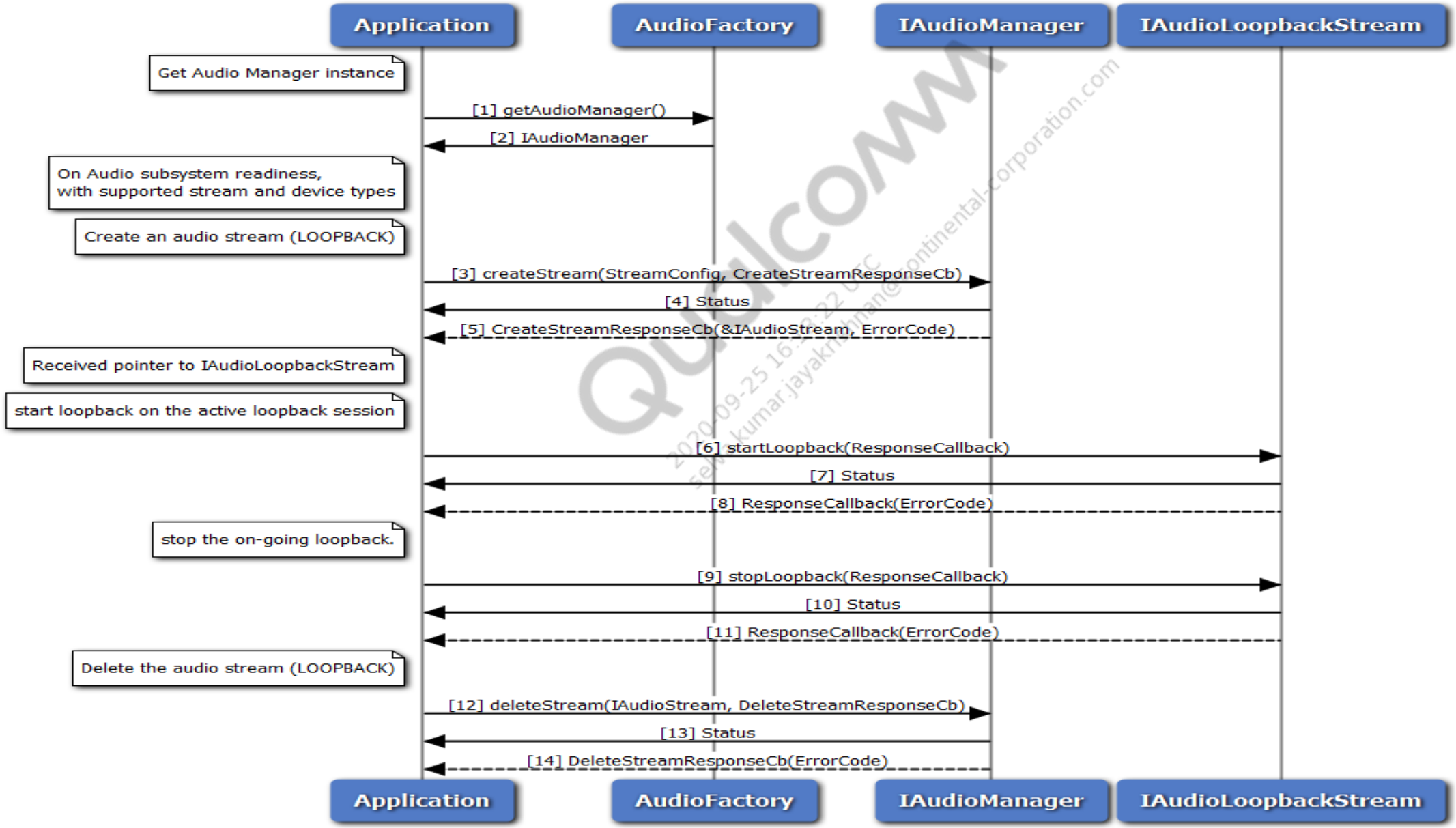
Loopback (Mic -> Speaker)



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TelSDK API

Documentation and Sample Apps

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TelSDK Voice and Audio Documentation

- Voice and Audio related documentation for Version 1.33.1 of the TelSDK

  https://developer.qualcomm.com/sites/default/files/docs/telematics/api/v1.33.1/a00009.html

- You can find the version of the TelSDK by running the "audio_console_app" on the target.
- There is also a table in Section 7 Additional Information of the Release Notes showing recent releases and which SDK version is used for each

```
adb shell
audio_console_app

Elapsed Time for Audio Subsystems to ready : 0.0055951s
-----------------------------------------------
            Audio console app - SDK v1.33.1
-----------------------------------------------
  1 - Voice Call
  2 - Playback
  3 - Capture
  4 - Loopback
  5 - Tone
  6 - TransCode

  ? / h - help
  q / 0 - exit
-----------------------------------------------
audio>
```

Date	Release tag	SA515M A7 TelSDK version	EAP TelSDK version
7/31/2020	r00075.1	v1.37.1	v1.37.1
6/18/2020	r00070.1	v1.34.0	v1.34.0
5/8/2020	r00060.1	v1.33.1	v1.33.1
3/9/2020	r00050.1	v1.30.1	v1.30.0

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Voice and Audio TelSDK Sample Applications

- Sample applications using the TelSDK API can be found at the following Open Source URLs:
  - voice_session_app - Sample App demonstration of Voice Stream

https://source.codeaurora.org/quic/le/platform/vendor/qcom-opensource/snaptel-sdk/tree/apps/samples/audio/voice_session_app?h=telsdk.lnx.2.0

  - playback_app - Sample App demonstration of Playback Stream

https://source.codeaurora.org/quic/le/platform/vendor/qcom-opensource/snaptel-sdk/tree/apps/samples/audio/playback_app/PlaybackApp.cpp?h=telsdk.lnx.2.0

  - capture_app - Sample App demonstration of Capture Stream

https://source.codeaurora.org/quic/le/platform/vendor/qcom-opensource/snaptel-sdk/tree/apps/samples/audio/capture_app?h=telsdk.lnx.2.0

  - audio_console_app – Sample console app for testing and demonstration

https://source.codeaurora.org/quic/le/platform/vendor/qcom-opensource/snaptel-sdk/tree/apps/tests/audio_console_app?h=telsdk.lnx.2.0

Note: The options in the test app may change, so our recommendation is to generate your own app for automating production tests as the TelSDK API will not change.

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TelSDK API

Debugging and Key Log Messages

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Logs for debugging

Telematics SDK logs
Before starting telsdk_console_app or audio_console_app, change the FILE_LOG_LEVEL parameter in “/etc/tel.conf” to “DEBUG”
i.e. FILE_LOG_LEVEL=DEBUG

QXDM logs:
File ... Manage Configuration (DMC)
Select "default.dmc", "Audio", "Voice"
<Apply Log Masks>

Logcat & Kernel logs:

```
adb root
adb shell mount -o rw,remount / (may be needed to make file system read/write)
adb shell
alias ls='ls --color=none’   (may be required for console)

echo -n "file msm-pcm-routing-v2.c +p" > /sys/kernel/debug/dynamic_debug/control
echo -n "file msm-pcm-voice-v2.c +p" > /sys/kernel/debug/dynamic_debug/control
echo -n "file msm-dai-q6-v2.c +p" > /sys/kernel/debug/dynamic_debug/control
echo -n “file msm-pcm-dtmf-v2.c +p” > /sys/kernel/debug/dynamic_debug/control
echo -n "file q6voice.c +p" > /sys/kernel/debug/dynamic_debug/control
echo -n “file q6asm.c +p” > /sys/kernel/debug/dynamic_debug/control
echo -n “file q6afe.c +p” > /sys/kernel/debug/dynamic_debug/control
echo -n “file sdx-target.c +p” > /sys/kernel/debug/dynamic_debug/control

logcat -v threadtime > /data/logcat.txt &
cat /proc/kmsg > /data/kmsg.txt &

<perform your test>

^C
adb pull /data/kmsg.txt .
adb pull /data/logcat.txt .
adb pull /data/vendor/telsdk/tel.log
```

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Key Log Messages - Critical Prerequisite

audio_console_app

```
09-07 20:45:01.062  1043  1043 V qahw_api: 1960:qahw_load_module start
01-06 00:00:23.805  1043  1043 D msm8974_platform: platform_init: ACDB initialized
01-06 00:00:23.916  1043  1043 D qahw    : qahw_load_module_l::Loaded HAL primary module 0xf39950
```

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Key Log Messages - Start Voice Call

```
audio> 1
voice> 1
[D] 1972/audio_console_app AudioManagerImpl.cpp(207) 3060707344: createStream
[D] 1972/audio_console_app AudioManagerImpl.cpp(594) 3023041600: onCreateStreamResponse stream type: 1 stream_ref: 0
01-06 00:01:15.343  1123  2813 V qahw_api: 1913:qahw_open_output_stream
01-06 00:01:15.349  1123  2813 V qahw_api: qahw_stream_open: end
voice> 9
[D] 1972/audio_console_app AudioStreamImpl.cpp(633) 3060707344: startAudio
[D] 1972/audio_console_app AudioStreamImpl.cpp(699) 2997863488: onStreamStartResponse
01-06 00:01:21.563  1123  2888 V qahw_api: 2380:qahw_stream_start start
01-06 00:01:21.580  1123  2888 D voice   : voice_start_usecase: enter usecase:voicemmode1-call
01-06 00:01:22.227  1123  2888 D voice   : voice_start_usecase: exit: status(0)
01-06 00:01:22.227  1123  2888 V qahw_api: 2401:qahw_stream_start end
<dial phone number and connect>
```

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Key Log Messages - Mute Voice path (Tx & Rx)

```
        voice> 8
Enter direction of stream: (0 for TX, 1 for RX) 0
Enter 0 to Unmute and 1 to Mute 1
[D] 1972/audio_console_app AudioStreamImpl.cpp(352) 3060707344: setMute
[D] 1972/audio_console_app AudioStreamImpl.cpp(371) 3060707344: setMute mute dir: 2
[D] 1972/audio_console_app AudioStreamImpl.cpp(425) 3006256192: onStreamSetMuteResponse
01-06 00:05:39.096  1123  3396 V qahw_api: 2681:qahw_stream_set_mute start
01-06 00:05:39.096  1123  3396 V qahw_api: qahw_get_device_mute_info mute_state device_mute=true;direction=tx
01-06 00:05:39.098  1123  3396 V qahw_api: 2706:qahw_stream_set_mute end
        voice> 8
Enter direction of stream: (0 for TX, 1 for RX) 1
Enter 0 to Unmute and 1 to Mute 1
[D] 1972/audio_console_app AudioStreamImpl.cpp(352) 3060707344: setMute
[D] 1972/audio_console_app AudioStreamImpl.cpp(371) 3060707344: setMute mute dir: 1
[D] 1972/audio_console_app AudioStreamImpl.cpp(425) 3006256192: onStreamSetMuteResponse
01-06 00:05:57.591  1123  3396 V qahw_api: 2681:qahw_stream_set_mute start
01-06 00:05:57.592  1123  3396 V qahw_api: qahw_get_device_mute_info mute_state device_mute=true;direction=rx
01-06 00:05:57.597  1123  3396 V qahw_api: 2706:qahw_stream_set_mute end
```

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Key Log Messages - Unmute Voice path (Tx & Rx)

```
        voice> 8
Enter direction of stream: (0 for TX, 1 for RX) 0
Enter 0 to Unmute and 1 to Mute 0
[D] 1972/audio_console_app AudioStreamImpl.cpp(352) 3060707344: setMute
[D] 1972/audio_console_app AudioStreamImpl.cpp(371) 3060707344: setMute mute dir: 2
[D] 1972/audio_console_app AudioStreamImpl.cpp(425) 3006256192: onStreamSetMuteResponse
01-06 00:05:39.096  1123  3396 V qahw_api: 2681:qahw_stream_set_mute start
01-06 00:05:39.096  1123  3396 V qahw_api: qahw_get_device_mute_info mute_state device_mute=false;direction=tx
01-06 00:05:39.098  1123  3396 V qahw_api: 2706:qahw_stream_set_mute end
        voice> 8
Enter direction of stream: (0 for TX, 1 for RX) 1
Enter 0 to Unmute and 1 to Mute 0
[D] 1972/audio_console_app AudioStreamImpl.cpp(352) 3060707344: setMute
[D] 1972/audio_console_app AudioStreamImpl.cpp(371) 3060707344: setMute mute dir: 1
[D] 1972/audio_console_app AudioStreamImpl.cpp(425) 3006256192: onStreamSetMuteResponse
01-06 00:05:57.591  1123  3396 V qahw_api: 2681:qahw_stream_set_mute start
01-06 00:05:57.592  1123  3396 V qahw_api: qahw_get_device_mute_info mute_state device_mute=false;direction=rx
01-06 00:05:57.597  1123  3396 V qahw_api: 2706:qahw_stream_set_mute end
```

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Key Log Messages - Stop Voice Call

<disconnect call>

voice> 10

```
[D] 1972/audio_console_app AudioStreamImpl.cpp(665) 3060707344: stopAudio
[D] 1972/audio_console_app AudioStreamImpl.cpp(716) 2997863488: onStreamStopResponse
01-06 00:01:51.840  1123  2888 V qahw_api: 2421:qahw_stream_stop start
01-06 00:01:52.083  1123  2888 V qahw_api: 2436:qahw_stream_stop end
```

voice> 2

```
[D] 1972/audio_console_app AudioManagerImpl.cpp(378) 3060707344: deleteStream stream type 1
[D] 1972/audio_console_app AudioManagerImpl.cpp(710) 3023041600: onDeleteStreamResponse Deleting stream_ref: 0
01-06 00:01:53.571  1123  2813 V qahw_api: 2325:qahw_stream_close start
01-06 00:01:53.573  1123  2813 V qahw_api: 2361:qahw_stream_close end
```

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Key Log Messages - PCM Playback

```
        audio> 2
        play> 1
        Please Select Audio Format : 1->PCM, 2->AMRWB+, 3->AMRNB, 4->AMRWB : 1
[D] 1972/audio_console_app AudioManagerImpl.cpp(207) 3060707344: createStream
[D] 1972/audio_console_app AudioManagerImpl.cpp(594) 3023041600: onCreateStreamResponse stream type: 2 stream_ref: 6
01-06 00:08:13.278  1123  2813 V qahw_api: 2082:qahw_stream_open start
01-06 00:08:13.279  1123  2813 D audio_hw_primary: adev_open_output_stream: Stream (0xb4605bf0) picks up usecase (deep-buffer-playback)
NOTE: may not be "deep-buffer-playback" (subject to change)
01-06 00:08:13.291  1123  2813 V qahw_api: qahw_stream_open: end
        play> 9
[D] 1972/audio_console_app AudioStreamImpl.cpp(871) 3035436096: write
01-06 00:08:14.995  1123  3453 V audio_hw_primary: start_output_stream: Opening PCM device card_id(0) device_id(0) format(0)
NOTE: May be different PCM device
01-06 00:08:15.306  1123  3453 D audio_hw_primary: start_output_stream: exit
01-06 00:08:15.306  1123  3453 V audio_hw_primary: out_write: frames=1024, frame_size=4, bytes_to_write=4096
NOTE: Ensure that you Delete Stream (Option 2) after playback is completed. There is no Stop Stream.
        play> 2
01-06 00:11:07.023   792   792 V qahw_api: 2307:qahw_stream_close start
01-06 00:11:07.023   792   792 D audio_hw_primary: adev_close_output_stream: enter:stream_handle(deep-buffer-playback)
01-06 00:11:07.058   792   792 V qahw_api: 2343:qahw_stream_close end
```

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Key Log Messages - AMR Playback

```
audio> 2
play> 1
Please Select Audio Format : 1->PCM, 2->AMRWB+, 3->AMRNB, 4->AMRWB : 2, 3 and 4
[D] 1972/audio_console_app AudioManagerImpl.cpp(207) 3060707344: createStream
[D] 1972/audio_console_app AudioManagerImpl.cpp(594) 3023041600: onCreateStreamResponse stream type: 2 stream_ref: 7
01-06 00:09:22.887  1123  2813 V qahw_api: 2082:qahw_stream_open start
01-06 00:09:22.888  1123  2813 D audio_hw_primary: adev_open_output_stream: Stream (0xb4605c50) picks up usecase (compress-offload-playback)
01-06 00:09:22.889  1123  2813 V qahw_api: qahw_stream_open: KV (music_offload_amrwbplus_bitstream_fmt=1;)
01-06 00:09:22.890  1123  2813 V qahw_api: qahw_stream_open: end
play> 9
[D] 1972/audio_console_app AudioStreamImpl.cpp(871) 2997863488: write
[D] 1972/audio_console_app QmiClient.hpp(266) 2997863488: sendRequest Client error(0) errStr: QMI_ERR_NONE Error Code: 0
onReadyForWrite should be received
Line 20289: [E] 1972/audio_console_app AudioQmiClient.cpp(1311) 2989470784: handleStreamWriteRespstream_write failed with response error: 38 (EXPECTED)
[D] 1972/audio_console_app AudioStreamImpl.cpp(914) 3007300672: onStreamWriteResponse
01-06 00:09:25.430  1123  3460 V audio_hw_primary: start_output_stream: Opening PCM device card_id(0) device_id(7) format(0)
01-06 00:09:25.765  1123  3460 D audio_hw_primary: start_output_stream: exit
01-06 00:09:25.770  1123  3460 V qahw_api: 1607:qahw_out_drain
01-06 00:09:34.815  1123  3460 V qahw_api: 1613:qahw_out_flush
NOTE: Ensure that you Delete Stream (Option 2) after playback is completed. There is no Stop Stream.
play> 2
01-06 00:11:07.023   792   792 V qahw_api: 2307:qahw_stream_close start
01-06 00:11:07.023   792   792 D audio_hw_primary: adev_close_output_stream: enter:stream_handle(compress-offload-playback)
01-06 00:11:07.058   792   792 V qahw_api: 2343:qahw_stream_close end
```

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Key Log Messages - DTMF Generation

```
<start voice call>
voice> 11
Enter the Gain : 10000
Enter the Low Frequency (697, 770, 852, 941) : 697
Enter the High Frequency (1209 1336 1477 1633) : 1209
Enter the duration (in ms (0-65534) and 65535 for infinite): 5000
[D] 1972/audio_console_app AudioStreamImpl.cpp(733) 3060707344: playDtmfTone
[D] 1972/audio_console_app AudioStreamImpl.cpp(804) 2997863488: onDtmfResponse
01-06 00:04:07.037  1123  3392 V qahw_api: 3079:qahw_stream_set_dtmf_gen_params kv set is
dtmf_tone_gain=5000;dtmf_low_freq=697;dtmf_high_freq=1209;dtmf_duration_ms=10000
```

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